

Interactive Dashboard & Tableau Storytelling

Q1. Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

Ans:

Tableau Dashboard

A Dashboard is a collection of multiple worksheets, charts, graphs, and filters displayed on a single screen.

Used to monitor and analyze data interactively in one view.

Focuses on real-time data exploration and decision-making.

Example: A hospital performance dashboard showing patient count, admissions, and bed occupancy.

Tableau Story Point

A Story Point is a sequence of dashboards or worksheets arranged to present a narrative or step-by-step analysis.

Used to explain insights, trends, and findings in a logical order.

Focuses on presenting a data-driven story to stakeholders.

Example: A healthcare report showing patient admissions, treatment outcomes, and recovery trends across multiple story points.

Healthcare-Related Use Cases

1. Hospital Performance Monitoring (Dashboard)

A hospital can use a Tableau Dashboard to track:

- Daily patient admissions
- Bed occupancy rates
- Emergency room visits
- Doctor availability

This helps administrators monitor hospital operations and make quick decisions.

2. Patient Treatment Journey Analysis (Story Point)

A healthcare organization can use Tableau Story Points to present:

- Patient registration statistics
- Diagnosis results
- Treatment progress
- Recovery and discharge outcomes

This allows management and medical staff to understand the complete patient journey and identify areas for improvement.

Q2. What are Actions in Tableau?

Explain how Filter Action can help hospital management analyze patient data.

Ans: **Actions** in Tableau are interactive features that allow users to perform specific operations by clicking, hovering, or

selecting marks in a visualization. Actions make dashboards more dynamic and help users explore data in greater detail.

A **Filter Action** allows hospital managers to click on a specific category and automatically filter related information across the dashboard.

Example:

Suppose a hospital dashboard contains:

- A chart showing departments (Cardiology, Neurology, Orthopedics, etc.)
- A table showing patient details
- A chart showing treatment outcomes

When a manager clicks on **Cardiology**, the Filter Action automatically updates the other charts and tables to display only Cardiology patients.

Benefits:

- Quickly analyze department-specific patient data.
- Identify departments with high patient loads.
- Monitor treatment success rates for a selected department.
- Improve resource allocation, such as beds, doctors, and equipment.
- Enable faster decision-making without manually applying filters.

Healthcare Use Case:

If the hospital administrator selects **Emergency Department**, the dashboard can instantly show the number of emergency patients, average waiting time, bed occupancy, and treatment outcomes for that department, helping management respond efficiently to operational needs.

Q9. Storytelling Flow Create a Tableau Story with a minimum of 4 Story Points:

Patient Distribution

Department Cost Analysis

Outcome Analysis

Doctor Performance

Ans: **Story Point 1: Patient Distribution**

Worksheet Used: Question 3

Visualization: Patient Count by Hospital Region

Observation:

- Shows how patients are distributed across different hospital regions.
- Identifies regions with the highest and lowest patient volumes.

Insight:

- Regions with higher patient counts may require additional medical resources and staff.

- Low-volume regions may indicate opportunities for outreach or service expansion.

Story Caption:

"Understanding where patients are coming from helps the hospital allocate resources more effectively and improve regional healthcare coverage."

Story Point 2: Department Cost Analysis

Worksheet Used: Question 4

Visualization: Treatment Cost by Department

Observation:

- Compares total treatment costs across departments.
- Highlights departments generating the highest healthcare expenditure.

Insight:

- High-cost departments require closer budget monitoring.
- Cost-intensive departments may need efficiency improvements without compromising patient care.

Story Caption:

"Analyzing departmental costs helps management identify major cost drivers and improve financial planning."

Story Point 3: Outcome Analysis

Worksheet Used: Question 6

Visualization: Patient Outcomes Distribution

Observation:

- Displays counts of outcomes such as Recovered, Under Treatment, Critical, etc.
- Provides a quick view of treatment effectiveness.

Insight:

- Higher recovery rates indicate successful healthcare delivery.
- Departments associated with lower recovery rates may require process improvements.

Story Caption:

"Patient outcomes provide a direct measure of healthcare quality and treatment effectiveness."

Story Point 4: Doctor Performance

Worksheet Used: Question 7

Visualization: Doctor Performance Analysis

Observation:

- Compares doctors using treatment cost and/or performance metrics.
- Identifies doctors handling higher patient loads or achieving stronger results.

Insight:

- Top-performing doctors can be recognized and used as benchmarks.
- Workload balancing can improve overall operational efficiency.

Story Caption:

"Evaluating doctor performance helps ensure quality care, fair workload distribution, and continuous improvement."

Q10. Business Insights

Based on your visuals, answer the following:

- 1. Which department generates the highest treatment cost?**
- 2. Which region has more critical patients?**
- 3. What business decisions can hospital management take from these insights?**

Ans: 1. Neurology

2. West

3. Optimize Neurology Department Costs

- Review treatment procedures and resource utilization.
- Allocate budget appropriately to support high-cost treatments.
- Invest in technology that improves efficiency without affecting patient care.

Strengthen Critical Care Services in West and South Regions

- Increase ICU beds, emergency facilities, and specialist availability.
- Deploy additional medical staff to manage the higher volume of critical patients.

Improve Resource Planning

- Use patient and cost trends to forecast staffing and equipment needs.
- Ensure departments with high treatment costs receive adequate funding.

Enhance Patient Outcomes

- Analyze causes of critical cases in high-risk regions.
- Introduce preventive healthcare programs and early diagnosis initiatives to reduce severe cases.